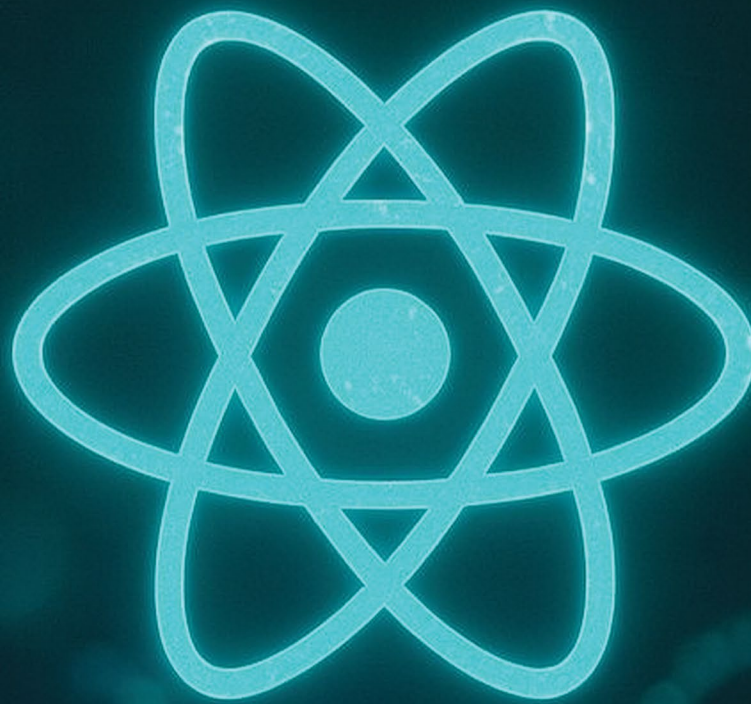


React Curriculum



COURSE CURRICULUM

1. Basics (Beginner Level)

Introduction to React

- o What is React? Why React? SPA concept
- o React vs Vanilla JS
- o Setup (Vite/CRA)

JSX

- o JSX rules & syntax
- o Embedding expressions { }
- o Fragments (<> </>)

Components

- o Function components
- o Component hierarchy
- o Props (passing data parent → child)
- o Default props & propTypes (validation)

State Basics

- o useState hook
- o Difference: Props vs State
- o Updating state correctly (immutability)

Event Handling

- o onClick, onChange, onSubmit
- o Passing functions as props

Rendering

- o Lists (.map())
- o Keys importance
- o Conditional rendering (if, ternary, &&)

COURSE CURRICULUM

2. Intermediate Level

Hooks (Core)

- o useEffect lifecycle methods (componentDidMount, etc.)
- o Cleanup functions

Forms

- o Controlled components (state-based)
- o Uncontrolled components (refs)
- o Multiple inputs, form validation

React Router

- o Setup with react-router-dom
- o Route, Link, Navigate
- o Nested routes
- o Protected routes

Context API

- o Avoiding prop drilling
- o useContext hook

API Integration

- o Fetch vs Axios
- o Loading & error handling
- o Displaying data in UI

Styling in React

- o CSS, CSS Modules
- o Styled Components
- o Tailwind CSS (most used now)

Lifting State Up

- o Sharing data between sibling components

COURSE CURRICULUM

3. Advanced Level

Advanced Hooks

- o useReducer (state management)
- o useRef (DOM refs, persist values)
- o useMemo (performance optimization)
- o useCallback (function memoization)

Custom Hooks

- o Reusable logic (useFetch, useForm)

Code Splitting

- o React.lazy + Suspense
- o Dynamic imports

Error Boundaries

- o Catch UI crashes

Performance Optimization

- o React.memo
- o Virtualization (react-window, react-virtualized)
- o Avoiding re-renders

Portals

- o Modals, tooltips outside root

React 18 Features

- o Concurrent rendering
- o useTransition
- o useDeferredValue

COURSE CURRICULUM

4. Expert Level

State Management

- o Redux Toolkit (modern Redux)
- o Zustand / Recoil / Jotai (alternative libraries)

Authentication

- o JWT, Cookies, Sessions
- o Protected routes with React Router

File & Folder Structure

- o Organizing components, hooks, utils

React Patterns

- o Higher Order Components (HOC)
- o Render Props
- o Compound Components pattern

React Suspense & Streaming

- o Suspense for data fetching
- o Concurrent features deep dive

SSR & Next.js

- o Next.js basics (pages, routing)
- o Server-side rendering
- o Static site generation

Testing

- o Jest basics
- o React Testing Library
- o Snapshot testing

Deployment

- o Build & deploy (Netlify, Vercel)
- o Environment variables



Scan the QR to know more